

Subject Description Form

Subject Code	COMP3235
Subject Title	Software Project Management
Credit Value	3
Level	3
Pre-requisite / Co-requisite / Exclusion	
Objectives	The objectives of this subject are to: 1. provide students a systematic approach to initiate, plan, execute, control and close a software project; 2. develop a good understanding of the nine project management areas, and the role of a typical project manager; 3. equip students with understanding of the best practices, and techniques used in project management processes; and 4. enable students to gain a good understanding of ISO and CMMI.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: <u>Professional/academic knowledge and skills</u> (a) appreciate the importance of software project management; (b) master the knowledge and skills for managing software projects; <u>Attributes for all-roundedness</u> (c) work together with others as a team; and (d) communicate both verbally and in writing software project information

Subject Synopsis/ Indicative Syllabus	Topic
	1. Project Management Fundamentals Attributes of project; project life cycle; project management processes; successful project manager; general management skills.
	2. Project Integration Management Project plan; change control; configuration management; corrective and preventive action.
	3. Project Scope Management Project charter; net present value; cost/benefit analysis; scope planning, definition, verification and change control.
	4. Project Time Management Project size and metrics; identifying activities; WBS; PBS; CPA; scheduling; critical chain.
	5. Project Cost Management Estimation techniques; earned value analysis; COCOMO; resource planning; value analysis; cost management plan, budgeting and control.
	6. Project Quality Management Quality model; quality definition; ISO; CMMI; improvement cycle; trend analysis.
	7. Human Resource Management Organisation structure; stakeholder analysis; team building; conflict; effective team; reward and recognition systems.
	8. Communication Management Communication means; communication techniques for teams of different sizes; barriers to communication; building effective team communication; reviews; performance reporting.
	9. Risk Management Different types of risk; risk response planning; risk analysis; risk monitoring and control.
	10. Procurement Management Procurement planning; source selection; contract administration; contract closeout; negotiation.
Teaching/ Learning Methodology	Lectures focus on introduction and explanation of key concepts and techniques. Tutorial and lab sessions provide students opportunity to practice the techniques and tools presented in class. Assignments and project allow students to deepen their understanding of the concepts taught in class and apply the theory and techniques to software process and project management. Students will be encouraged to work in groups to share and present ideas, review other's work, and develop teamwork skill.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed			
			a	b	c	d
	Continuous Assessment	30%				
	1. Project				✓	✓
	2. Quizzes		✓	✓		
	Examination	70%	✓	✓		
	Total	100%				
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Quizzes act as a measure on the understandings of the students on the basic concepts of the software project management. Project will be used to measure the understandings of the students about the current practice in process and project management. The students can improve their presentation and communication skills through the project presentation, and practice teamwork. Students can also develop their analytic and problem-solving skills. Examination will be used as an overall measure of the understandings of the students on software project management.						
Student Study Effort Expected	Class contact:					
	▪ Lecture				36 Hrs.	
	▪ Tutorial/Lab				3 Hrs.	
	Other student study effort:					
	▪ Work on project, Self-Study				66 Hrs.	
	Total student study effort				105 Hrs.	
Reading List and References	Textbook:					
	1. Schwalbe, Kathy, Information Technology Project Management (9th Edition), Cengage Learning, 2018.					
	Reference Books:					
	1. A Guide to the Project Management Body of Knowledge (6th Edition), Project Management Institute, 2017.					
	2. Hughes, B., Cotterell, M., Software Project Management (5th Edition), McGraw-Hill, 2009.					